Finding Thurstone: modeling Comparative Judgment data with `R` (and `Stan`)

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Abstract

The Classical Comparative Judgment (CCJ) analysis has become the standard approach for analyzing comparative judgment (CJ) data across multiple fields. Researchers favor CCJ because it offers a relatively simple approach for trait measurement and follow-up analyses. This simplicity stems from three features: reliance on restricted Thurstone-based models to estimate latent traits, limited consideration of relevant assessment design features, and the separation of trait measurement from follow-up analysis. Recent studies, however, question whether these assumptions adequately reflect contemporary CJ applications and whether omitting relevant design features, together with the separation of measurement and analysis, may bias estimates and reduce inferential precision.

To address these concerns, Rivera et al. (2025) proposed the Information-Theoretical Comparative Judgment (ITCJ) analysis, an umbrella analytical workflow designed to accommodate the complexity observed in applied CJ settings. The approach enables researchers to specify models tailored to the assumed CJ data-generating process, thereby reducing reliance on simplifying assumptions. In addition, by integrating measurement, analysis, and relevant assessment design features within a single framework, the approach also supports coherent estimation and inference. However, these potential advantages require empirical validation.

Therefore, this study evaluates the ITCJ analysis using a synthetic speech-quality dataset and benchmarks its performance against the CCJ approach. In line with principles of open and reproducible science, the study also provides step-by-step guidance on data simulation, prior specification, model estimation, and interpretation using `R`, `Stan`, `cmdstan`, and `brms`. Ultimately, by combining empirical validation with a fully reproducible implementation, the study equips researchers with practical tools for applying the ITCJ approach to complex CJ assessments.

Keywords: tutorial, causal inference, bayesian inference, thustone, comparative judgement

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1. Introduction

Comparative judgment (CJ) has emerged as a valuable methodology for measuring latent traits across diverse fields, including education (Kimbell, 2012; Jones and Inglis, 2015; van Daal et al., 2016; Bartholomew et al., 2018), psychology (Maydeu-Olivares and Böckenholt, 2005), political sciences (ZuccoJr. et al., 2019), linguistics (Boonen et al., 2020), and criminology (Seymour et al., 2022; Seymour and Hernandez, 2025). In CJ studies, judges actively compare pairs of stimuli to determine which stimulus exhibits more of the latent trait of interest (Thurstone, 1927b,a).

The *Classical Comparative Judgment* (CCJ) analysis has become the standard approach for analyzing CJ data across multiple fields, including education, linguistics, and psychology (see, for example, Wu, 2025; Thwaites and Paquot, 2024; Maydeu-Olivares and Böckenholt, 2005). Researchers favor this approach because it provides a convenient and simple method for measuring traits and conducting follow-up analyses (Andrich, 1978; Pollitt, 2012b). Three features underpin its simplicity. First, it relies on restricted Thurstone-based models to estimate latent traits. These models facilitate estimation by imposing simplifying assumptions about the traits, judges, and stimuli involved in the CJ assessment under study (Thurstone, 1927a; Bramley, 2008). Second, it pays limited attention to relevant CJ assessment design features that shape the data-generating process, such as the sampling and comparison mechanisms and potential judge biases (Rivera et al., 2025). Third, the approach separates trait estimation from follow-up analysis (Pollitt, 2012b), simplifying the latter by focusing only on mean trait estimates or residuals obtained from the former process (Rivera et al., 2025).

Recent studies, however, question whether the assumptions of these models hold in contemporary CJ applications, and whether the limited consideration of relevant assessment design features, together with the separation of trait estimation and analysis, may bias estimates and reduce the precision of inferences. (Bramley, 2008; Kelly et al., 2022; Rivera et al., 2025).

To address these concerns, Rivera et al. (2025) proposed the *Information-Theoretical Comparative Judgment* (ITCJ) analysis, an umbrella analytical workflow designed to accommodate the complexity often observed in applied CJ settings. The approach leverages causal and Bayesian inference methods to combine Thurstone's core theoretical principles with relevant CJ assessment design features. This integration enables researchers to develop models tailored to the assumed data-generating process of the CJ system under study, thereby reducing reliance on simplifying assumptions. Moreover, by integrating measurement, analysis, and relevant design features within a single framework, the ITCJ analysis facilitates a single, coherent and flexible approach for trait estimation and statistical inference.

The flexibility of the ITCJ analysis is especially valuable in education, linguistics, and psychology,

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where CJ data often arise from complex assessment designs (Bramley, 2008; van Daal et al., 2017; Kelly et al., 2022) or involve comparisons of complex, heterogeneous stimuli (van Daal et al., 2016; Lesterhuis et al., 2018; Chambers and Cunningham, 2022). Its relevance, however, extends beyond these fields to any discipline using CJ assessments, as the approach allows researchers to model the processes they aim to study rather than those constrained by restrictive statistical models.

1.1. Research goals

The ITCJ analysis holds theoretical promise for producing reliable trait estimates and accurate statistical inferences (Rivera et al., 2025). However, as the authors acknowledge, this promise remains empirically untested. This study addresses this gap by *empirically validating the ITCJ analysis* using a synthetic speech-quality dataset. Specifically, it evaluates the accuracy and reliability of the resulting trait estimates and inferential parameters relative to those obtained under the standard CCJ analysis.

Moreover, in line with principles of open and reproducible science, the study also provides step-by-step guidance on data simulation, prior specification, model estimation, and interpretation using `R` (R Core Team, 2015), `Stan` (Stan Development Team., 2026b,a), and the interface packages `cmdstanr` (Gabry et al., 2025) and `brms` (Bürkner, 2017, 2018). Notably, by combining empirical validation with a fully reproducible implementation, the study equips researchers with practical tools needed to apply the the ITCJ approach to more complex CJ assessments.

The remainder of this manuscript is organized into five sections. Section 2 reviews the standard analytical approach applied to CJ data: the CCJ analysis, and compares it with the ITCJ analysis. Section 3 describes the assumed data-generating process for the simulated dataset, the simulation procedure, the practical implementation of each analytical approach, and the evaluation criteria aligned with the research goals. Section 4 presents the data description and modeling results. Section 5 interprets the findings, outlines future research directions, and considers the study limitations. Finally, Section 6 offers the concluding remarks.

2. A tale of two analytical approaches

Pairwise comparison data are typically analyzed using the standard *CCJ analysis*. However, the *ITCJ analysis* proposed by Rivera and colleagues (2025) offers an alternative approach. This section provides an overview of both.

2.1. The CCJ analysis

The CCJ analysis typically proceeds through two sequential steps, each serving a distinct purpose: first, estimating traits, and second, conducting follow-up analyses (Pollitt, 2012a,b; Jones et al., 2019; Boonen et al., 2020; Chambers and Cunningham, 2022; Bouwer et al., 2023). Trait estimation generally relies on Thurstone-based models, and yield two primary outputs: mean trait estimates and comparison-level residuals. For example, Boonen et al. (2020) used the Bradley-Terry-Luce (BTL) model (Bradley and Terry, 1952; Luce, 1959) to derive mean latent scores representing children's overall speech quality, together with the corresponding comparison-level residuals.

Researchers then use these mean trait estimates and comparison-level residuals to conduct follow-up analyses. On the one hand, analyses of the mean latent scores fulfill several purposes, including summarizing estimates to aid interpretation, aggregating stimulus-level estimates to the individual level, partitioning variability between and within individuals, and drawing inferences about factors that influence trait scores. For example, Maydeu-Olivares and Böckenholt (2005) applied various factor models to latent means that represent individuals' preferences for compact cars, to provide a more parsimonious representation of these preferences. Similarly, Boonen et al. (2020) applied a multilevel regression model to mean latent estimates to examine whether children's age or hearing status affects their speech quality scores.

On the other hand, analyzing the residuals typically seeks to aggregate the remaining variability at the judge level, partition residual variability between and within judges, test for systematic biases, and identify potential misfitting stimuli, individuals, or judges. For instance, Wu (2025) fitted an analysis of variance (ANOVA) model to judges' infit statistics to examine whether judge expertise influences judgment behavior. The infit statistics–a weighted average of the squared Pearson residuals–indicate the extent to which judges' decisions deviate from expectations relative to those of other judges (Wright and Masters, 1982).

This two-step CCJ analysis has become a standard practice in many CJ applications across education, linguistics, and psychology (see, e.g., Wu, 2025; Thwaites and Paquot, 2024; Maydeu-Olivares and Böckenholt, 2005). However, the approach has three key limitations: first, it relies on restricted Thurstone-based models to estimate traits; second, it often gives limited consideration to relevant CJ assessment design features; and third, it separates trait estimation from follow-up analysis.

Regarding the first limitation, although restricted Thurstone-based models facilitate trait estimation by imposing simplifying assumptions about traits, judges, and stimuli (Thurstone, 1927a; Bramley, 2008), violations of these assumptions can undermine the reliability and validity of the resulting trait estimates (Perron and Gillespie, 2015; Rivera et al., 2025). This concern is exemplified by Wu (2025), who, under the BTL model, implicitly treats second-language English texts as independent stimuli that are equally informative about individuals' writing performance, despite their heterogeneous and complex nature (Thurstone, 1927b; Andrich, 1978; Bramley, 2008; Kelly et al., 2022). Notably, if the texts differ in informativeness due to this heterogeneous nature, the BTL assumption of equal dispersions may be violated, potentially leading to an overestimation of the trait reliability and misleading conclusions about differences between stimuli (McElreath, 2020; Wu et al., 2022). Moreover, if the characteristics of these texts unevenly influence judges' perceptions, systematic biases may arise. Such biases can induce dependencies among stimuli, further contributing to an over- or underestimation of the trait reliability (van der Linden, 2017; Ackerman, 1989; Hoyle, 2023).

Concerning the second limitation, while giving limited attention to specific CJ assessment design features may also simplify trait estimation or follow-up analyses, Rivera and colleagues (2025) argued that neglecting elements that define the CJ data-generating process can obscure their broader role within the assessment system. The authors illustrated this concern by examining the sampling and comparison mechanisms. They showed that explicit inclusion of these mechanisms in the conceptual CJ model reveal their role as sources of missing data. They further argued that clarifying their function supports a more comprehensive evaluation of their impact on trait estimation, statistical inference, and the design of more complex assessment setups.

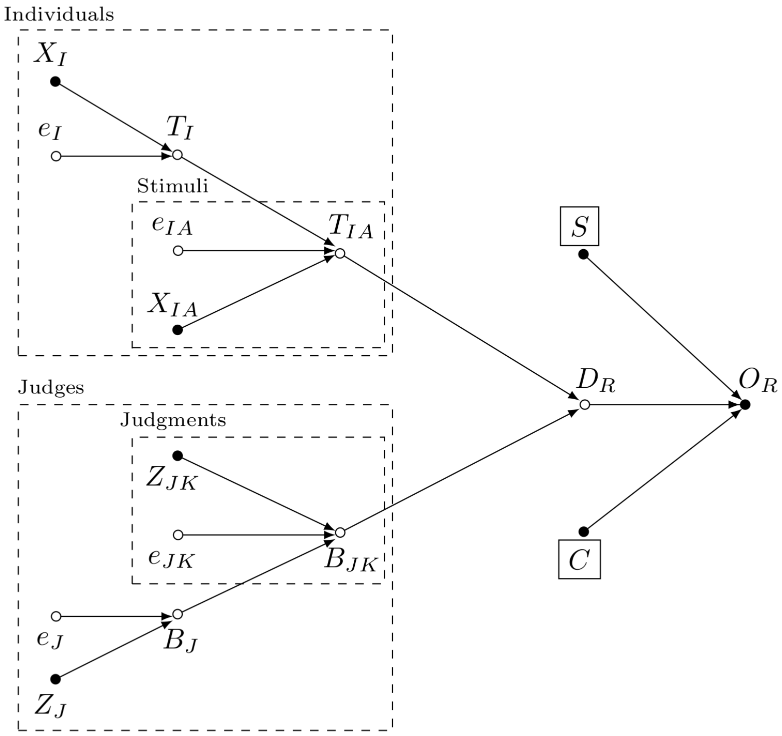
Lastly, concerning the third limitation, although separating trait estimation from follow-up analysis can simplify the latter, focusing solely on mean trait estimates or residuals obtained from the estimation stage inadvertently treats these outputs as fixed values rather than as uncertain parameter estimates (McElreath, 2020; Rivera et al., 2025). This separation prevents the propagation of uncertainty from the estimation to the analysis stage (Pritikin, 2020), which can bias parameter estimates and reduce their precision. The direction and magnitude of these biases are often difficult to predict. Effects may be attenuated, inflated, or remain unchanged depending on the level of uncertainty in the scores and the true effects under investigation (McElreath, 2020; Kline, 2023; Hoyle, 2023). In addition, reduced precision lowers statistical power and increases the risk of Type I and Type II errors (McElreath, 2020).

Several studies have made valuable attempts to incorporate specific features that address some of the limitations discussed above. However, most studies propose incremental models that address one or two issues at a time. As a result, other relevant components of the CJ process remain unrepresented, which may lead to model misspecification and produce biased or imprecise parameter estimates (Everitt and Skrondal, 2010). For example, Seymour et al. (2022) and Seymour and Hernandez (2025) extended the BTL model to incorporate spatially correlated stimuli and tied outcomes, applying these models to measure area-level deprivation and honour-based abuse, respectively. Although these extensions captured important aspects of the CJ data, the analysis did not explicitly model other potentially relevant factors, such as the sampling mechanism associated with their "convenience" data sample, possible judge biases and judge-specific effects, or hierarchical structures expected in the data, such as wards nested within counties. Similar patterns of underrepresented features appear in Pritikin (2020), Schramm and Batchelder (2021), and Gray et al. (2024), where models typically address one or two key design features or sources of heterogeneity at a time.

Notably, the *ITCJ analysis* proposed by Rivera and colleagues (2025) provides an alternative approach that addresses the limitations outlined above. In the following section, we summarize this approach, with particular emphasis on its differences from the CCJ analysis.

2.2. The ITCJ analysis

In contrast to the CCJ approach, the ITCJ analysis does not begin with the application of a predefined statistical model to the CJ data, from which primary outputs are extracted for subsequent analysis. Instead, it starts by explicitly representing the relationships among observed and latent variables that are thought to characterize the data-generating process of the CJ system under study. This structure is represented using a *Directed Acyclic Graph (DAG)* and an associated *Structural Causal Model (SCM)* (Morgan and Winship, 2014; Gross et al., 2018; Neal, 2020). On the basis of this causal representation, a *probabilistic model* tailored to the system under study is derived, from which one or more *statistical models* are subsequently formulated. These models are then used to estimate, summarize, and conduct statistical inferences on the traits of interest.



(a) DAG

$$
\begin{aligned}
O_R &:= f_O(D_R, S, C) \\
D_R &:= f_D(T_{IA}, B_{JK}) \\
T_{IA} &:= f_T(T_I, X_{IA}, e_{IA}) \\
T_I &:= f_T(X_I, e_I) \\
B_{JK} &:= f_B(B_J, Z_{JK}, e_{JK}) \\
B_J &:= f_B(Z_J, e_J)
\end{aligned}
$$

$$
\begin{aligned}
e_I &\perp \{e_J, e_{IA}, e_{JK}\} \\
e_J &\perp \{e_{IA}, e_{JK}\} \\
e_{IA} &\perp e_{JK}
\end{aligned}
$$

(b) SCM

Figure 1: ITCJ analysis, plausible data-generating process for a CJ-based speech quality study

As an example, Figure 1 presents a DAG and SCM representing a plausible data-generating process for a CJ-based speech quality assessment study. In DAG 1a, closed circles represent observed variables, whereas open circles represent latent (unobserved) variables. Arrows indicate the direction of the assumed influences between variables. Dashed boxes denote the assumed hierarchical structure of the data, such as stimuli nested within individuals. In contrast, SCM 1b represents the same structure using a more formal mathematical notation, in which functions $f_{(\cdot)}$, known as *structural equations*, represent variables as non-parametric functions of other observed or latent variables. These equations use the symbol ':=' to denote the asymmetric direction of the assumed influences, and the symbol ' $\perp$ ' to represent *d-separation*, a concept closely related to statistical (conditional) independence. Rivera et al. (2025) provides a more detailed explanation of these concepts.

Figure 1 uses X_I to denote the set of variables believed to influence the individual traits T_I, i.e., the individuals' ability to produce high-quality speech. Examples of such individual-level variables include hearing status and age–the latter serving as a proxy for language development (Boonen et al., 2020). Similarly, X_{IA} denotes the set of variables assumed to influence the stimulus traits T_{IA}, which represent the latent quality of the stimulus under comparison. Examples of such stimulus-level variables include the length of the speech sample and whether the sample originates from spontaneous speech or imitation. Notably, the stimulus traits T_{IA} correspond to what the CJ literature refers to as *discriminal processes*, which are though to capture the psychological effect a stimulus exerts on the judge, or more simply, the judge's perception of the stimulus quality (Thurstone, 1927b,a).

In a similar fashion, Z_J denotes the set of variables believed to influence the judges' bias B_J, that is, the systematic deviation of a judge's perception of a stimulus from the consensus. Examples of such judges-level variables include speech-judgment expertise and prior training in the CJ task. Similarly, Z_{JK} denotes the set of variables assumed to influence the judgment-level biases B_{JK}, which represent systematic deviations of an individual judgment from the judge-specific bias. Examples of such judgment-level variables include the time of day or day of the week at which the comparison was made, as well as the number of comparisons completed by a judge, used as a proxy for learning or fatigue effects during the CJ task. The error terms e_I, e_{IA}, e_J, and e_{JK} capture residual variation at each level.

Notably, because individual and stimulus traits are, in essence, not directly observable, Thurstone (1927b; 1927a) introduced the *Law of Comparative Judgment*. This law posits that, in pairwise comparisons, a judge perceives the stimulus with a *discriminal process* positioned further along the trait continuum as possessing more of the trait (Bramley, 2008). This suggests that pairwise comparison outcomes depend on relative, rather than absolute, positions on the continuum, implying that the observed CJ outcome is determined by the difference between the underlying discriminal processes (traits) of the stimuli (Rivera et al., 2025). Moreover, because discriminal processes become observable only through judges' perceptions–and these perceptions may involve bias or systematic deviations from consensus (see Pollitt and Elliott, 2003; van Daal et al., 2016)–judge- and judgment-level biases are appropriately treated as integral components of the CJ system from the outset (Rivera et al., 2025). In this representation, these relationships are captured by the arrows from the T_{IA} and B_{JK} vectors to the *discriminal difference* D_R vector, and the corresponding structural equation $D_R := f_D(T_{IA}, B_{JK})$.

Furthermore, Figure 1a represents the sampling and comparison mechanisms as two conditioned variables that determine how population outcomes become the “observed” sample outcomes (S and C, respectively shown with surrounding squares to denote conditioning). In this particular representation, these variables are depicted without incoming arrows, indicating that they are assumed to be independent of all other variables in the DAG. This, in turn, implies that S corresponds to a *simple random sampling (SRSg)* design, in which each repeated judgment, judge, stimulus, and individual has an equal probability of inclusion within their respective groups (Lawson, 2015), while C corresponds to a *random allocation comparative design* (Bramley, 2015), also known as an *incomplete block design* (Lawson, 2015), in which each judgment has an equal probability of being included in the sample.

Finally, Figure 1a shows that the observed CJ outcome O_R is jointly influenced by the discriminal difference D_R and the sampling and comparison mechanisms (S and C, respectively). In typical CJ applications, O_R is often modeled as following a Bernoulli distribution (see Pollitt, 2012b; Bramley, 2008; Boonen et al., 2020; Seymour and Hernandez, 2025). However, the current DAG is intentionally agnostic to this specific distributional choice, thereby accommodating alternative outcome distributions as special cases within the ITCJ approach. This flexibility is particularly relevant in settings where the comparison task yields non-binary outcomes, for which extensions beyond the Bernoulli model may be more appropriate (Rivera et al., 2025). Examples of such outcomes can be found in Agresti (1992), and more recently in Pritikin (2020) and Lingel et al. (2024).

$$
\begin{aligned}
&P(O_R, D_R, S, C, T_{IA}, X_{IA}, e_{IA}, T_I, X_I, e_I, B_{JK}, Z_{JK}, e_{JK}, B_J, Z_J, e_J) \\
&\quad = P(O_R \mid D_R, S, C) \cdot P(S) \cdot P(C) \\
&\qquad \cdot P(D_R \mid T_{IA}, B_{JK}) \\
&\qquad \cdot P(T_{IA} \mid T_I, X_{IA}, e_{IA}) \cdot P(X_{IA}) \\
&\qquad \cdot P(T_I \mid X_I, e_I) \cdot P(X_I) \\
&\qquad \cdot P(B_{JK} \mid B_J, Z_{JK}, e_{JK}) \cdot P(Z_{JK}) \\
&\qquad \cdot P(B_J \mid Z_J, e_J) \cdot P(Z_J) \\
&\qquad \cdot P(e_I) P(e_{IA}) P(e_J) P(e_{JK})
\end{aligned}
$$

Figure 2: Probabilistic model, joint distribution (left hand side) factorized into conditional probability distributions (CPDs, right hand side).

Once researchers specify the SCM characterizing the assumed CJ data-generating process, as in Figure 1b, they can translate it into a probabilistic model that factorizes the joint distribution of the variables into a product of simpler *conditional probability distributions (CPDs)* (Pearl et al., 2016; Neal, 2020), as illustrated in Figure 2. From this formulation, they can then derive one or more statistical models tailored to the CJ system under study (see Section 3.4.2), enabling coherent trait estimation and inference within a unified framework.

At this point, it becomes clear that the ITCJ approach differs fundamentally from the CCJ approach in its treatment of trait estimation, summarization, and inference. This difference enables it to address the three main limitations of the latter. First, although the ITCJ analysis still relies on Thurstone-based models for trait estimation, these models are no longer treated as pre-specified simplifying abstractions; instead, they are adapted to the specific data-generating process of the CJ system under study, allowing researchers to explicitly model the processes of substantive interest. Second, the approach explicitly incorporates relevant CJ assessment design features, including the sampling and comparison mechanisms as well as judge and judgment biases. Third, the ITCJ analysis no longer separates trait estimation from follow-up analyses, but rather integrates both within a unified and systematic approach (Rivera et al., 2025). This integration enables the propagation of uncertainty across all levels of the assumed hierarchical structure of the data.

Building on this framework, the following section details the methods used to implement and evaluate the ITCJ analysis against the CCJ approach.

3. Methods

To address its goal of empirical model validation using a synthetic speech-quality dataset, this study follows the *Bayesian research workflow* (Depaoli and van de Schoot, 2017; Pearl and Mackenzie, 2018; Neal, 2020; Gelman et al., 2020; Schad et al., 2020; Betancourt, 2020; McElreath, 2024a,b), depicted in Figure 3. This workflow guides the construction of the synthetic dataset, as well as the specification, estimation, and evaluation of the analysis models.

Notably, as shown in the figure, the workflow components form an interconnected and partly non-linear process, reflecting the inherent complexity of statistical modeling. To ensure clarity and reproducibility, the remainder of this article unpacks each stage step-by-step.

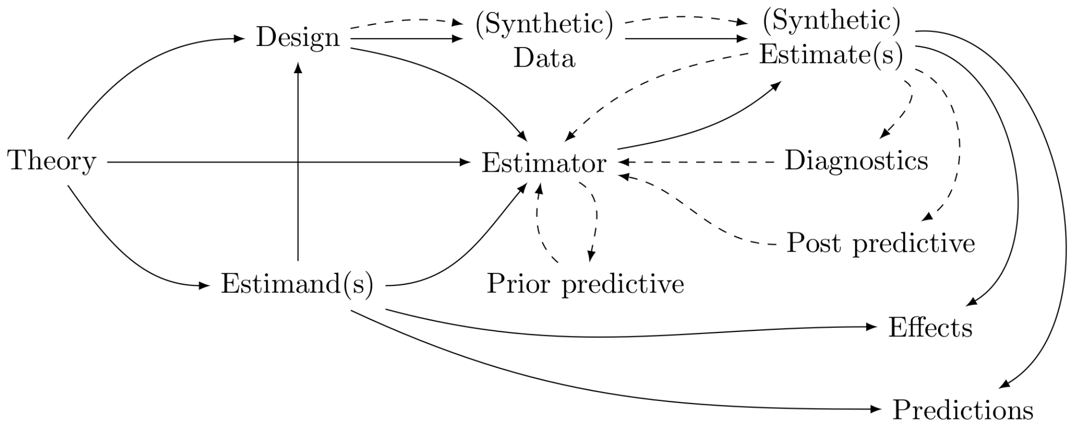


Figure 3: Bayesian research workflow.

3.1. From Theory to Design: Data-generating assumptions

Because the study aims to conduct empirical validation, it employs a synthetic dataset. Synthetic data enables rigorous evaluation of model performance under controlled conditions, where both the underlying data-generating mechanisms and the target parameters are known by design.

In this study, the data-generating assumptions draw on key features and findings reported in Boonen et al. (2020). Specifically, it maintains the author's original goal–namely, to examine whether children's age and hearing status influence overall speech quality–while implementing this within a CJ task that preserves the core design features and empirical findings of the original study.

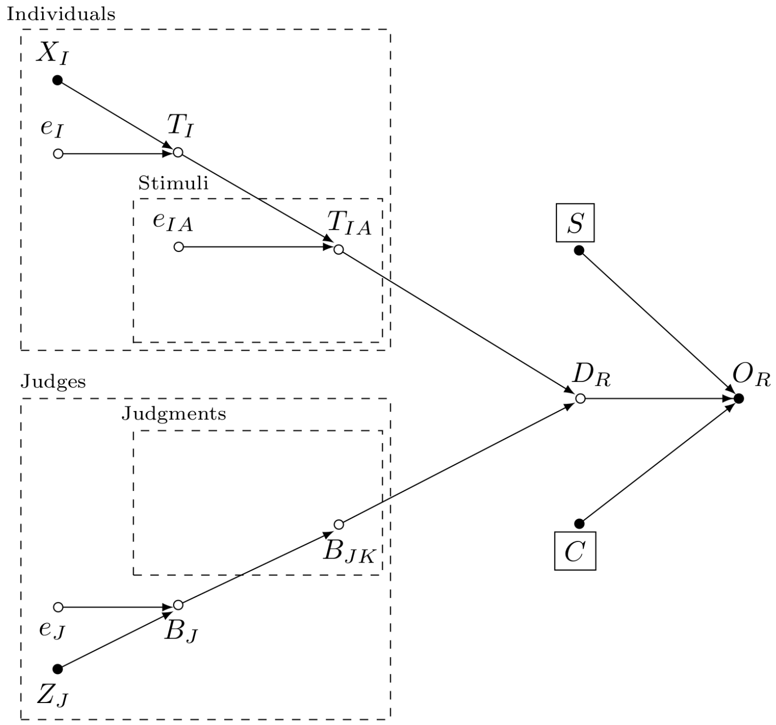


Figure 4: DAG of the assumed data-generating process underlying the synthetic conceptual population

Figure 4 presents the DAG incorporating the design features of Boonen et al. (2020). It depicts stimuli nested within individuals who vary in characteristics X_I, such as age and hearing status. Notably, in contrast to Figure 1a, the model assumes no additional stimulus-level attributes, as reflected by the absence of the variable X_{IA}. Moreover, the DAG includes judges who differ in speech-judgment expertise Z_J and who provide a single judgment per stimulus pair, as indicated by the absence of variables Z_{JK} and e_{JK} previously shown in Figure 1a.

Lastly, similar to Figure 1a, the DAG represents the sampling and comparison mechanisms (S and C, respectively) without incoming arrows, indicating their independence from all other variables in the model. This structure implies that the model draws stimuli, individuals, and judges from a conceptual population using SRSg algorithms (Lawson, 2015) and assigns judges to comparisons using random allocation comparative designs (Bramley, 2015), respectively.

$$O_R := f_O(D_R, S, C)$$
$$D_R := f_D(T_{IA}, B_{JK})$$
$$T_{IA} := f_T(T_I, e_{IA})$$
$$T_I := f_T(X_I, e_I)$$
$$B_J := f_B(Z_J, e_J)$$

$$e_{IA} \perp \{e_I, e_J\}$$
$$e_I \perp \{e_J\}$$

(a) SCM

$$P(O_R \mid D_R, S, C)$$
$$P(D_R \mid T_{IA}, B_{JK})$$
$$P(T_{IA} \mid T_I, e_{IA})$$
$$P(T_I \mid X_I, e_I)$$
$$P(B_J \mid Z_J, e_J)$$

$$P(e_{IA})P(e_I)P(e_J)$$

(b) Probabilistic model

$$O_R \overset{iid}{\sim} \text{Bernoulli}\left[\text{inv\_logit}(D_R)\right]$$
$$D_R = (T_{IA}[i,a] - T_{IA}[h,b]) + B_{JK}[j,k]$$
$$T_{IA} = T_I + e_{IA}$$
$$T_I = \beta_{XI} X_I + e_I$$
$$B_J = \beta_{ZJ} Z_J + e_J$$

$$e \sim \text{Multi-Normal}(\mu, \Sigma)$$
$$\Sigma = VQV$$

(c) Statistical model

Figure 5: SCM, Probabilistic and Statistical model of the "True" Data-Generating Process Underlying the Synthetic Conceptual Population

Next, the study incorporates Boonen and colleagues' (2020) empirical findings by specifying a statistical model derived from the probabilistic model implied by the system's SCM. This procedure is depicted in Figure 5. For this purpose, the model first organizes individual-level predictors into the vector $X_I = \{XIc, XId[1], XId[2], XId[3]\}$ with corresponding effect vector $\beta_{XI} = \{\beta_{XIc}, \beta_{XId[1]}, \beta_{XId[2]}, \beta_{XId[3]}\}$, and judge-level variables into the vector $Z_J = \{ZJd[1], ZJd[2], ZJd[3]\}$ with corresponding effect vector $\beta_{ZJ} = \{\beta_{ZJd[1]}, \beta_{ZJd[2]}, \beta_{ZJd[3]}\}$.

The model then assumes that each one-unit increase in age (XIc) leads to a 0.1-logit increase in speech quality for children aged five to nine, as captured by $\beta_{XIc} = 0.1$. It further encodes a strict ordering of effects across three hearing-status groups, such that normal-hearing children ($NH; XId[1]$) exhibit higher speech quality than hearing-impaired children with hearing aids ($HI - HA; XId[2]$), who in turn outperform hearing-impaired children with cochlear implants ($HI - CI; XId[3]$). This ordering is represented by $\beta_{XId[1]} > \beta_{XId[2]} > \beta_{XId[3]}$, where $\beta_{XId[1]} = 1$, $\beta_{XId[2]} = 0$, and $\beta_{XId[3]} = -1$, corresponding to a one-logit difference between adjacent groups.

Moreover, the model assumes that expertise with the judgment task does not systematically influence average judgment bias. Consequently, audiologists ($AU; ZJd[1]$), primary teachers ($PT; ZJd[2]$), and inexperienced listeners ($IL; ZJd[3]$) all have zero mean effects, such that $\beta_{ZJd[1]} = \beta_{ZJd[2]} = \beta_{ZJd[3]} = 0$.

Similarly, to incorporate assumptions about the covariance structure of the error terms, the model collects them in the vector $e = \{e_I, e_{IA}, e_J\}$. It then assumes that this vector follows a Multivariate Normal distribution with mean $\mu = [0, 0, 0]^T$ and covariance matrix Σ. Moreover, it specifies the covariance matrix via a standard deviation–correlation decomposition (McElreath, 2020), such that

$\Sigma = VQV$, where V is a diagonal matrix of standard deviations and Q is a correlation matrix. More specifically, the study defines the matrices Q and V as follows:

$$Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; V = \begin{bmatrix} s_{XI} & 0 & 0 \\ 0 & s_A & 0 \\ 0 & 0 & s_{ZJ} \end{bmatrix}$$

Figure 6: Assumptions for the standard deviation-correlation decomposition of the covariance matrix Σ.

The Q matrix indicates that the generating model assumes the three types of error terms are mutually independent. This implies that residual variability in stimulus traits is uncorrelated with residual variability in individual traits, and that both are uncorrelated with residual variability in judges' biases.

Moreover, the model incorporates additional assumptions through the V matrix. Specifically, it assumes that the speech-quality trait variability differs across hearing-status groups. In particular, NH children exhibit higher variability ($s_{XId[1]} = 1.5$) than $HI-HA$ and $HI-CI$ children, both of whom are assumed to show lower and equal variability ($s_{XId[2]} = s_{XId[3]} = 0.75$). Considering these specifications, and assuming an equal-probability SRSg design, the model implies an average between-individual variability of $s_{XI} = \sum_{g=1}^{3} s_{XId[g]}/3 = 1$, which exceeds the assumed within-individual (stimulus-level) variability $s_A = 0.2$.

Finally, although the original study found no evidence that judges' expertise influences biases in their speech-quality perceptions, it did not examine whether expertise affects the *variability* of those biases. To assess the ability of the ITCJ approach to investigate such hypothesis, the generating model assumes that audiologists (AU) exhibit less bias variability than primary teachers (PT), who in turn exhibit less bias variability than inexperienced listeners (IL). This ordering is represented by $s_{ZJd[1]} < s_{ZJd[2]} < s_{ZJd[3]}$, where $s_{ZJd[1]} = 0.5$, $s_{ZJd[2]} = 1$, and $s_{ZJd[3]} = 1.5$. Given these specifications, and an equal-probability SRSg design, the model implies an average between-judge bias variability of $s_{ZJ} = \sum_{g=1}^{3} s_{ZJd[g]}/3 = 1$.

3.2. From Design to Data: Data simulation

3.3. From Theory to Estimands: Target parameters

3.4. From Estimator to Estimate(s): The models

3.4.1. The CCJ analysis

3.4.2. The ITCJ analysis

3.4.2.1. Model 1.

3.4.2.2. Model 2.

3.4.2.3. Model 3.

3.4.2.4. Model 4.

3.4.2.5. Model 5.

3.4.2.6. Model 6.

3.5. From Estimate(s) to Diagnostics and Posterior predictives: The evaluation criteria

4. Results

4.1. Data description

4.2. Data modeling

4.2.1. The CCJ analysis

4.2.2. The ITCJ analysis

4.2.2.1. Model 1.

4.2.2.2. Model 2.

4.2.2.3. Model 3.

4.2.2.4. Model 4.

4.2.2.5. Model 5.

4.2.2.6. Model 6.

4.2.2.7. Model comparison.

5. Discussion

5.1. Future research directions

5.2. Study limitations

6. Conclusion

Declarations

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7. Appendix

7.1. Appendix A: Stationarity, converge and mixing

7.2. Appendix B: Misfit observations

7.3. Appendix C: Sample size calculations

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